# THE UNIVERSITY OF AUCKLAND

**SEMESTER ONE 2025**
**Campus: City**

**COMPUTER SCIENCE**

**COMPSCI 713: AI Fundamentals**

**MID-SEMESTER TEST**
**29h of April 2025**

**Time Allowed: 60 minutes (5 minutes reading time + 55 min answering time)**

**NOTES:**

- This test is out of **15** marks.
- There are 6 short-answer questions. Attempt **ALL** questions.
- This is a restricted book test. You are only allowed to use a **double-sided page of handwritten notes**.
- Digital devices, including calculators and phones are not permitted.
- Write your answers directly after each question.
- Cross out notes and other work you do not wish to be assessed.
- We privilege quality over quantity, i.e., you do not need to write very long answers. Be concise and clear.
- Write as clearly as possible, in plain English.
- Overflow space is available at the end of the test paper, indicate at the end of the original question if you are using overflow space.

| Question | Out of | Marks |
|---|---|---|
| 1 | 2 | |
| 2 | 3 | |
| 3 | 2 | |
| 4 | 2 | |
| 5 | 3 | |
| 6 | 3 | |
| **TOTAL** | **15** | |

| **Name:** | **UPI:** | **ID:** |
|---|---|---|

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**Answer ALL questions.**

**Question 1 [2 marks]**

This question tests your understanding of two fundamental types of symbolic logic: *propositional logic* – which deals with complete statements and logical connectives; and *first-order logic* – which involves quantifiers and reasoning about individuals in a domain. Answer both part (a) and part (b).

(a) Consider the example below.

> In a smart office, the alarm system is governed by the following rule:
> "If the door is open or the motion sensor is triggered, then the alarm will sound."
>
> Let the propositions be:
> P: The door is open
> Q: The motion sensor is triggered
> R: The alarm sounds
>
> The rule is logically expressed as:
> $$(P \vee Q) \rightarrow R$$
> Today, you observed that the alarm did not sound.

Use propositional logic to deduce what must be true about P and Q. Show your steps clearly. **[1 mark]**

(b) Consider the example below.

At a university, the disciplinary policy states:
"Every student who cheats in the exam will be disqualified from the course."

Let:
Cheat(x): student x cheats
Disqualified(x): student x is disqualified

This rule is represented in first-order logic as:

$$\forall x\ (\text{Cheat}(x) \rightarrow \text{Disqualified}(x))$$

You are told:
Student Alice is not disqualified.

Based on the rule and the observed fact, use first-order logic to determine whether Alice cheated. Show your reasoning. **[1 mark]**

**Question 2 [3 marks]**

In a Logic Neural Network (LNN), each logical formula is associated with a lower bound (L) and an upper bound (U), which indicate how confident the model is about the truth of the formula. Consider the example below in answer part (a) and part (b).

> An autonomous vehicle is using an LNN to decide whether to trigger a collision alert based on two observed conditions:
> - P: "The object is very close" ($L_P$ = 0.8, $U_P$ = 0.9)
> - Q: "The object is moving fast" ($L_Q$ = 0.3, $U_Q$= 0.6)
>
> The LNN rule for triggering an alert is:
> Alert ← P ∨ Q
>
> The system uses a classification threshold of α = 0.7:
> - If L ≥ α → definitely true
> - If U < α → definitely false
> - If L < α < U → uncertain

(a) Based on the threshold α = 0.7, is the alert status: **[1 mark]**
A. definitely true
B. definitely false
C. uncertain
(Choose one; no proof is required.)

(b) Why is using bounds (instead of just a single probability) beneficial in safety-critical applications such as autonomous driving? **[2 marks]**

**Question 3 [2 marks]**

(a) Briefly explain how a translation-based embedding model such as TransE learns representations for entities and relations in a knowledge graph. **[1 mark]**

(b) What is the basic idea behind TransE's scoring function? **[1 mark]**

**Question 4 [2 marks]**

CART is a recursive algorithm for decision tree induction driven by impurity reduction.

What exactly is meant by saying CART is ‘greedy’?

**Question 5 [3 marks]**

Consider a rule that uses the attributes STRONG and HEAVY that assesses suitability of an individual for the Olympic sport of hammer throwing:

IF STRONG AND HEAVY THEN HAMMER_THROWER

Contrast how the above rule might work using traditional logic as compared to Fuzzy Logic.

**Question 6 [3 marks]**

You've been asked to program a genetic algorithm (GA) to train the controller for the legs of a walking robot like Big Dog.

Your inputs will include the desired speed and direction of the robot, as well as the target height (e.g., standing tall, or crouched low). The algorithm will have access to the speed, direction, height and angle (i.e., 'attitude' – pitch, yaw and roll) of the robot's body from the prior time step.

Name the elements that should be part of the fitness function for the GA. You don't need to write a complete equation; just describe the key terms that would contribute to the sum of the fitness score.

END OF TEST QUESTIONS – OVERFLOW PAGES FOLLOW

OVERFLOW PAGES